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TEST FIXTURE FOR ACCELERATED TESTING OF DIFFERENT LOADS

Abstract

A test fixture for accelerated testing of different loads includes a fixture shaft adapted to be detachably connected to a test shaft. A pulley is fixedly supported by the fixture shaft. A cable has a first end fixedly attached to the pulley and a second end detachably attached to a test mass.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present disclosure relates generally to a test fixture and, more particularly, to a test fixture for a rotating shaft or gear train that allows for rapid changes to test different loads as well as a varying load within a given load cycle.

BACKGROUND

[0002] In product development, there are often several different material and design concepts for a particular mechanism, for example, without limitation, an electric door motor gear assembly used to open and close the entry doors of a school bus. The different design concepts need to be tested to determine which is the most desirable or robust. In the past, various bench testing techniques have been used to apply fixed loads to a rotating shaft or gear train over various cycles and durations. However, it is sometimes necessary to perform accelerated testing to quickly quantify the differences among different design concepts.

[0003] A need therefore exists for a test fixture that allows for rapid changes to applied test loads as well the ability to vary the applied load within a given load cycle. It would be beneficial if the test load could be quickly changed to achieve a different but constant motor or gear load. It would be further beneficial if the load applied in a load cycle could be easily varied, for example without limitation, to add an increased load near the beginning or near the end of the load cycle. It would be further beneficial if the test fixture is relatively compact and adaptable to many gear designs.

SUMMARY OF THE INVENTION

[0004] In one aspect of the invention, a test fixture for accelerated testing comprises a fixture shaft adapted to be detachably connected to a test shaft. A pulley is fixedly supported by the fixture shaft. A cable has a first end fixedly attached to the pulley and a second end detachably attached to a test mass.

[0005] In another aspect of the invention, a test fixture for accelerated testing comprises a fixture shaft adapted to be connected to a test shaft. A pulley is fixedly supported by the fixture shaft. A cable has a first end fixedly attached to the pulley and a second end detachably attached to a test mass of a plurality of distinct test masses, and the fixture shaft includes bearings disposed to support rotation of the fixture shaft.

[0006] In a further aspect of the invention, a test fixture for accelerated testing comprises a fixture shaft adapted to be connected to a test shaft. A pulley is fixedly supported by the fixture shaft, and bearings are disposed on the fixture shaft to support rotation of the fixture shaft. A cable has a first end fixedly attached to the pulley and a second end detachably attached to a test mass of a plurality of distinct test masses. A spring is attached to a fixed surface disposed between the test mass and the pulley and configured so that when rotation of the pulley wraps the cable to lift the test mass, the spring is in the path of the test mass.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The foregoing and other features of the present disclosure will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. These drawings depict only several exemplary embodiments in accordance with the disclosure and are, therefore, not to be considered limiting of the scope of the present disclosure.

[0008] FIG. **1** is a first schematic view of a test fixture according to an exemplary embodiment.

[0009] FIG. **2** is a second schematic view of the test fixture of FIG. **1**.

[0010] FIG. **3** illustrates the steps of an exemplary method for using the test fixture of FIGS. **1** and **2**.

[0011] FIG. **4** is a first schematic view of a test fixture according to another exemplary embodiment.

[0012] FIG. **5** is a second schematic view of the test fixture of FIG. **4**.

[0013] FIG. **6** illustrates the steps of an exemplary method for using the test fixture of FIGS. **4** and **5**.

[0014] In the following detailed description, various exemplary embodiments are described with reference to the appended drawings. The skilled person will understand that the accompanying drawings are schematic and simplified for clarity. Like reference numerals refer to like elements or components throughout. Like elements or components will therefore not necessarily be described in detail with respect to each figure.

DETAILED DESCRIPTION

[0015] Referring to FIGS. **1** and **2**, two views of an exemplary embodiment of test fixture **100** for accelerated testing is shown. In accordance with an exemplary embodiment, test fixture **100** comprises fixture shaft **110** that is adapted to be detachably connected to test shaft **120**. In an exemplary embodiment, coupling **130** is used to make the detachable connection. For example, without limitation, coupling **130** is a beam shaft or rigid shaft coupling.

[0016] Pulley **140** is shown edge-on in FIG. **1** and along an axis of rotation in FIG. **2** and is preferably fixedly supported by fixture shaft **110**, so that a rotation of fixture shaft **110** results in an equal rotation of pulley **140**. In an exemplary embodiment, cable or wire **150** has a first end **160** that is fixedly attached to pulley **140**. For example without limitation, first end **160** is fastened into a groove **170** of pulley **140** by a fastener like a screw or bolt **180**, or first end **160** can be welded or otherwise fixedly attached to pulley **140**. In an exemplary embodiment, rotation of pulley **140** results in cable or wire **150** being wrapped onto or unwrapped from pulley **140**. In an exemplary embodiment, second end **190** of cable or wire **150** is detachably attached to test mass **200**.

[0017] In an exemplary embodiment as shown in FIG. **1**, test mass **200** is one of a plurality of test masses **200**, **200**A, **200**B, etc., that detachably attach to second end **190**, wherein each of the plurality of test masses **200**, **200**A, **200**B, etc., has a distinct mass, illustrated in FIG. **1** for example without limitation, as the test masses **200**, **200**A, **200**B, **200**C, **200**D, and **200**E, where each different reference numeral is representative of having a distinct mass. In any exemplary embodiment, there can be as many or as few of the plurality of test masses **200**, **200**A, **200**B, etc. For example, **200**, **200**A, **200**B, . . . , through **200**Z, as are desired for quick and easy changing of the load applied. In an exemplary embodiment, test masses **200**, **200**A, **200**B, etc., are incremented in weight by a fixed amount and/or are labeled by their weight for easy recognition by a technician. In an exemplary embodiment, the test masses **200**, **200**A, **200**B, etc., detachably attach to second end **190** by a hook on second end **190** attaching around a bar or loop on test mass **200**, or by a hook extending from test mass **200** attaching around a loop in second end **190**, or by a quick release latch, or by any mechanism for detachable connection as is known in the art.

[0018] In an exemplary embodiment, test fixture **100** further comprises bearings **210** disposed on fixture shaft **110** to support rotation of fixture shaft **110**. For example, without limitation as shown in FIG. **1**, bearings **210** can in turn be supported by support arms **220** extending from bench or tabletop **230**. Without being held to theory, bearings **210** allow for test masses **200**, **200**A, **200**B, etc., to be supported by pulley **140** without causing an unwanted bending stress in fixture shaft **110** across pulley **140**.

[0019] Still referring to FIGS. **1** and **2**, in an exemplary embodiment test fixture **100** further comprises at least one of an upper counter switch or contact **240** and a lower counter switch or contact **250** supported below pulley **140**. For example, in an exemplary embodiment upper and lower counter switches or contacts **240** and **250** can be physically supported on member **260** that extends from bench or tabletop **230**. In an exemplary embodiment, each of test masses **200**, **200**A, **200**B, etc., further comprise counter arm **270** that is configured to make contact with each of the upper and lower counter contacts **240** and **250**, or close each of the upper and lower contact switches **240** and **250** upon contact. In an exemplary embodiment, each of the upper and lower counter switches or contacts **240** and **250** is connected, either wirelessly or via a wired connection **280**, to supporting computer hardware, software, and data storage memory **290** (shown schematically as a cloud **290**) for recording the number of contacts/switch closures.

[0020] Referring to FIG. **2**, in an exemplary embodiment, test shaft **120** is connected to gear train

300 or other structure or mechanism **300** to which a test load is to be applied. In operation, gear train **300** is preferably driven as it would be in actual use, for example by driving motor (not shown) in first and second directions which causes test shaft **120** together with fixture shaft **110** to rotate in first and second opposite directions, which, in turn, rotates pulley **140** in first and second opposite directions, thus raising and lowering test mass **200**.

[0021] Referring now to FIG. **3**, in an exemplary embodiment, a method **400** for testing a gear train **300** or other structure or mechanism **300** begins with step **410** of providing test fixture **100** comprising upper counter contact **240** and lower counter contact **250**. At step **420**, gear train **300** is connected to test shaft **120**, which is preferably coupled to fixture shaft **110**, for example via coupler **130**. At step **430**, gear train **300** is first driven to turn test shaft **120** in a first direction of rotation to raise test mass **200** until counter arm **270** contacts upper counter contact **240** or closes upper counter switch **240**. At step **440**, gear train **300** is second driven to turn test shaft **120** in a second direction of rotation to lower test mass **200** until counter arm **270** contacts lower counter contact **250** or closes lower counter switch **250**. At step **450**, first and second driving steps **430** and **440** are repeated a predetermined number of times as counted by upper and lower counter contacts (or switches) **240**, **250** or until gear train **300** fails.

[0022] Referring now to FIGS. **4** and **5**, an exemplary embodiment of test fixture **500** is similar to test fixture **100** described hereinabove with regard to FIGS. **1** and **2**. All of the same structures present in test fixture **500** are labeled with the same reference numerals as were used in the description of test fixture **100** above. However, text fixture **500** includes additional structure not present in test fixture **100**, as is described fully hereinbelow.

[0023] Whereas test fixture **100** can be used to provide a constant load on gear train **300** or other mechanism to be tested **300**, the exemplary embodiment of test fixture **500** allows for an applied load to be varied through the load cycle. In an exemplary embodiment, test fixture **500** further comprises spring **510** or pair of springs **510** attached to fixed surface **520** that is disposed between test mass **200** and pulley **140**. For example, without limitation, in an exemplary embodiment, spring or springs **510** are detachably attached at first end **530** to bottom surface **520** of bench or tabletop **230**. Spring or springs **510** are configured to compress vertically in FIGS. **4** and **5**, so that when rotation of pulley **140** wraps wire or cable **150** to lift test mass **200**, spring or springs **510** are in the path of test mass **200**. As test mass **200** rises, spring or springs **510** get compressed. Without being held to theory, the force, F that results from a compressed spring varies linearly with the distance of compression X. Mathematically, this is usually written as $F=-kX$, where k is commonly known as the spring constant. In effect, the presence of spring or springs **510** causes the force applied by hanging mass **200** to increase by an amount equal to the compression force of spring or springs **510**, which allows each load cycle to have a variable load.

[0024] Still referring to FIGS. **4** and **5**, the exemplary embodiment of test fixture **500** also comprises upper and lower counter switches or contacts **240**, **250** supported below fixed surface **520**. Test mass **200** further comprises counter arm **270** configured to make contact with each of upper and lower counter contacts **240**, **250**, or close each of the upper and lower upon contact switches **240**, **250** upon contact. However, in the embodiment of test fixture **500** upper counter contact or switch **240** is disposed closer to fixed surface **520** than second end **540** of spring or springs **510** so that test mass **200** compresses spring or springs **510** when counter arm **270** makes contact with upper counter contact or switch **240**.

[0025] Referring now to FIG. **6**, in an exemplary embodiment, method **600** for testing gear train **300** or other structure or mechanism **300** begins with step **610** of providing test fixture **500** comprising upper counter contact **240** and lower counter contact **250**. At step **620**, gear train **300** is connected to test shaft **120**, which is coupled to fixture shaft **110**, for example via coupler **130**. At step **630**, gear train **300** is first driven to turn test shaft **120** in a first direction of rotation to raise test mass **200** until counter arm **270** contacts upper counter contact **240** or closes upper counter switch **240**. At step **640**, gear train **300** is second driven to turn test shaft **120** in a second direction of rotation to lower test mass **200** until counter arm **270** contacts lower counter contact **250** or closes lower counter switch **250**. It should be noted that in the exemplary method **600**, step **630** also preferably compresses spring or springs **510**, which adds additional load to the end of first driving step **630** and to the beginning of second driving step **640**. At step **650**, first and second driving steps **430** and **440** are repeated a predetermined number of times as counted by upper and lower counter contacts (or switches) **240**, **250** or until gear train **300** fails. Other methods for using test fixtures **100**, **500** include executing one of the methods **400**, **600** for a predetermined number of cycles, changing test mass **200**, repeating the executed method **400**, **600**, changing test mass **200**, etc., wherein test mass **200** is changed as many times as is desired, a predetermined number of times, or until gear train **300** fails.

[0026] It should be noted that, in an accordance with an exemplary embodiment, additional controlling processors, memory, and communications hardware and software are present within cloud **290** for controlling the number of predetermined cycles and the duration of each cycle applied to the test fixture **100**, **500** in the methods **400**, **600** and other methods as described above, and for storing data including the number of cycles executed, as well as stresses on gear train **300** or other mechanism being tested **300** as measured by strain gauges or other stress-measuring tools.

[0027] With respect to the use of plural and/or singular terms herein, those having skill in the art can translate from the plural to the singular and/or from the singular to the plural as is appropriate to the context and/or application. The various singular/plural permutations may be expressly set forth herein for sake of clarity.

INDUSTRIAL APPLICABILITY

[0028] A test fixture for load testing gear trains and driven rotating shafts is presented, which allows for rapid changes to applied test loads as well the ability to vary the applied load within a given load cycle. The test fixture is relatively compact and adaptable to many gear designs. The test fixture can be manufactured in industry for use by consumers and manufacturers of machinery having rotating shafts and driven gear trains.

[0029] Numerous modifications to the present invention will be apparent to those skilled in the art in view of the foregoing description. It is not desired to limit the invention to the exact construction and operation shown and described and, accordingly, all suitable modifications and equivalents may be resorted to falling within the scope of the invention. Accordingly, this description is to be construed as illustrative only of the principles of the invention and is presented for the purpose of enabling those skilled in the art to make and use the invention and to teach the best mode of carrying out same. The exclusive rights to all modifications which come within the scope of the appended claims are reserved. All patents, patent publications and applications, and other references cited herein are incorporated by reference herein in their entirety.

Claims

1. A test fixture for accelerated testing comprising: a fixture shaft adapted to be detachably connected to a test shaft; a pulley fixedly supported by the fixture shaft; and a cable having a first end fixedly attached to the pulley and a second end detachably attached to a test mass.

2. The test fixture of claim 1, further comprising bearings disposed on the fixture shaft to support rotation of the fixture shaft.

3. The test fixture of claim 1, further comprising a plurality of additional test masses that detachably attach to the second end, wherein each of the plurality of additional test masses has a distinct mass.

4. The test fixture of claim 1, wherein the test shaft is connected to a gear train.

5. The test fixture of claim 1, further comprising at least one of an upper counter contact and a lower counter contact supported below the pulley, wherein the test mass further comprises a counter arm configured to make contact with the at least one of the upper counter contact and the lower counter contact.

6. A method for load testing a gear train, the method comprising the steps of: providing the test fixture of claim **5** comprising the upper counter contact and the lower counter contact; connecting a gear train to the test shaft; coupling the test shaft to the fixture shaft; first driving the gear train to turn the test shaft in a first direction of rotation to raise the test mass until the counter arm contacts the upper counter contact; second driving the gear train to turn the test shaft in a second direction of rotation to lower the test mass until the counter arm contacts the lower counter contact; and repeating the first and second driving steps a predetermined number of times as counted by the upper counter contact and the lower counter contact or until the gear train fails.

7. The test fixture of claim 1, further comprising a spring attached to a fixed surface disposed between the test mass and the pulley and configured so that when rotation of the pulley wraps the cable to lift the test mass, the spring is in the path of the test mass.

8. The test fixture of claim 7, wherein an upper counter contact and a lower counter contact are supported below the fixed surface, and wherein the test mass further comprises a counter arm configured to make contact with the upper counter contact and the lower counter contact.

9. The test fixture of claim 8, wherein the upper counter contact is disposed closer to the fixed surface than the spring so that the test mass compresses the spring when the counter arm makes contact with the upper counter contact.

10. A test fixture for accelerated testing comprising: a fixture shaft adapted to be connected to a test shaft; a pulley fixedly supported by the fixture shaft; a cable having a first end fixedly attached to the pulley and a second end detachably attached to a test mass; and wherein the fixture shaft includes bearings disposed to support rotation of the fixture shaft.

11. The test fixture of claim 10, wherein the test shaft is connected to a gear train.

12. The test fixture of claim 10, further comprising a spring attached to a fixed surface disposed between the test mass and the pulley and configured so that when rotation of the pulley wraps the cable to lift the test mass; the spring is in the path of the test mass.

13. The test fixture of claim 12, further comprising at least one of an upper counter contact and a lower counter contact supported below the fixed surface, wherein the test mass further comprises a counter arm configured to make contact with the at least one of the upper counter contact and the lower counter contact.

14. The test fixture of claim 13, wherein the upper counter contact is disposed closer to the fixed surface than the spring so that the test mass compresses the spring when the counter arm makes contact with the upper counter contact.

15. A method for load testing a gear train, the method comprising the steps of: providing the test fixture of claim **14** comprising the upper counter contact and the lower counter contact; connecting the gear train to the test shaft; coupling the test shaft to the fixture shaft; first driving the gear train to turn the test shaft in a first direction of rotation to raise the test mass until the counter arm contacts the upper counter contact; second driving the gear train to turn the test shaft in a second direction of rotation to lower the test mass until the counter arm contacts the lower counter contact; and repeating the first and second driving steps a predetermined number of times as counted by the upper counter contact and the lower counter contact or until the gear train fails.

16. A test fixture for accelerated testing, the fixture comprising: a fixture shaft adapted to be connected to a test shaft; a pulley fixedly supported by the fixture shaft; bearings disposed on the fixture shaft to support rotation of the fixture shaft; a cable having a first end fixedly attached to the pulley and a second end detachably attached to a test mass; and a spring attached to a fixed surface disposed between the test mass and the pulley and configured so that when rotation of the pulley wraps the cable to lift the test mass, the spring is in the path of the test mass.

17. The test fixture of claim 16, wherein the test shaft is coupled to a gear train.

18. The test fixture of claim 16, further comprising at least one of an upper counter contact and a lower counter contact supported below the fixed surface, wherein the test mass further comprises a counter arm configured to make contact with the upper counter contact and the lower counter contact.

19. The test fixture of claim 18, wherein the upper counter contact is disposed closer to the fixed surface than the spring so that the test mass compresses the spring when the counter arm makes contact with the upper counter contact.

20. The test fixture of claim 18 comprising the upper counter contact and the lower counter contact.